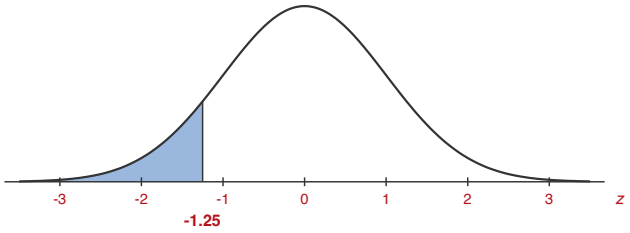
Answers: Standard Normal Areas Practice

Each answer shows the table lookups. A spreadsheet gives the same answers, sometimes with the last digit off by 1: NormalAreasPractice-Answers.xlsx.

Example: Using the Table

Find the left tail area for $z = -1.25$.



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611

Split $z = -1.25$ into the **row** -1.2 (the ones digit and first decimal place) and the **column** .05 (the second decimal place). Where they meet:

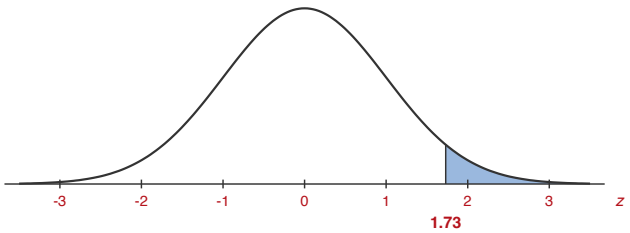
$$\text{left tail area} = 0.1056 \approx 11\%$$

Our table has only **left tail areas**. For the other areas, look up left tail areas first:

- **Right tail area** $= 1.0 -$ left tail area
- **Area between** z_1 and z_2 = (left tail area for z_2) $-$ (left tail area for z_1)

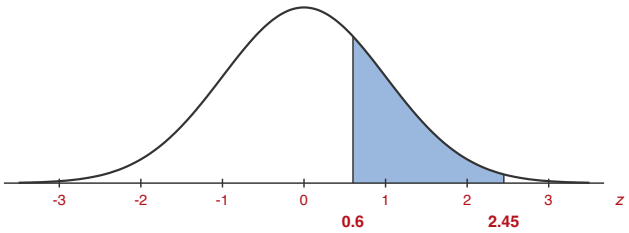
Problems

1. Find the right tail area for $z = 1.73$



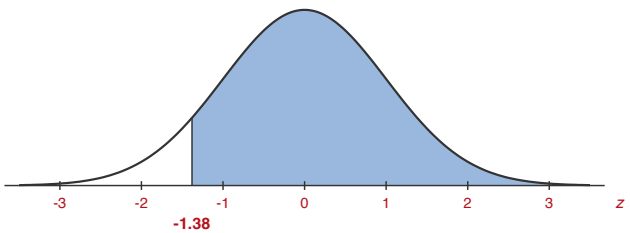
Row 1.7, column .03: left tail area $= 0.9582$
Right tail area $= 1.0 - 0.9582 = 0.0418$

2. Find the area between $z = 0.6$ and $z = 2.45$



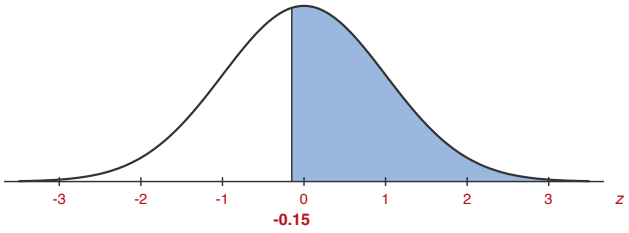
$z = 2.45$: row 2.4, column .05 gives 0.9929
$z = 0.6$: row 0.6, column .00 gives 0.7257
Area between $= 0.9929 - 0.7257 = 0.2672$

3. Find the area to the right of $z = -1.38$



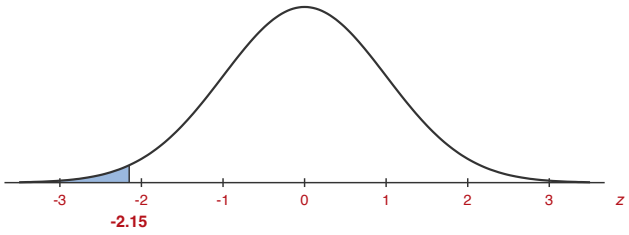
Row -1.3, column .08: left tail area $= 0.0838$
Right tail area $= 1.0 - 0.0838 = 0.9162$

4. Find the area where $z > -0.15$



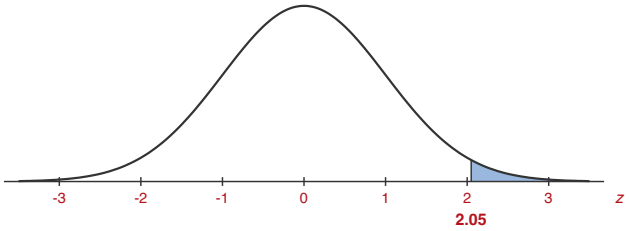
Row -0.1, column .05: left tail area $= 0.4404$
Right tail area $= 1.0 - 0.4404 = 0.5596$

5. Find the left tail area for $z = -2.15$



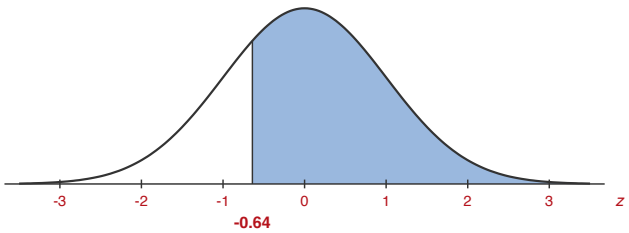
Row -2.1, column .05: left tail area $= 0.0158$

6. Find the area for z greater than 2.05



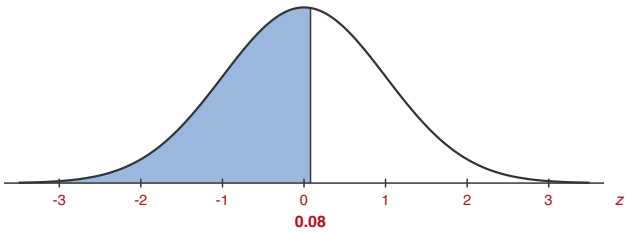
Row 2.0, column .05: left tail area $= 0.9798$
Right tail area $= 1.0 - 0.9798 = 0.0202$

7. Find the right tail area for $z = -0.64$



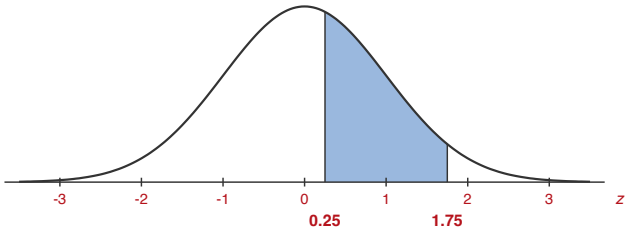
Row -0.6, column .04: left tail area $= 0.2611$
Right tail area $= 1.0 - 0.2611 = 0.7389$

8. Find the area to the left of $z = 0.08$



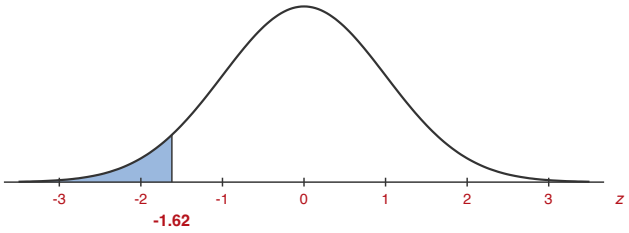
Row 0.0, column .08: left tail area $= 0.5319$

9. Find the area where $0.25 < z < 1.75$



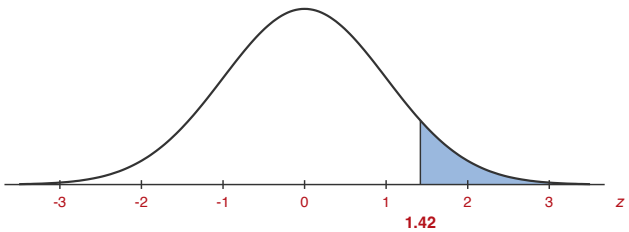
$z = 1.75$: row 1.7, column .05 gives 0.9599
$z = 0.25$: row 0.2, column .05 gives 0.5987
Area between $= 0.9599 - 0.5987 = 0.3612$

10. Find the area where $z < -1.62$



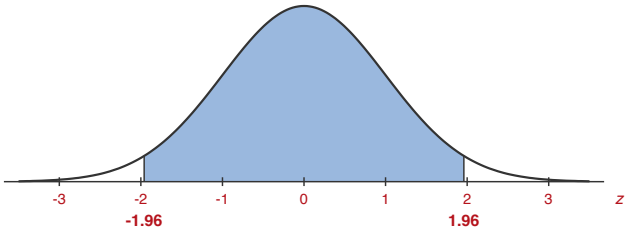
Row -1.6, column .02: left tail area $= 0.0526$

11. Find the area to the right of $z = 1.42$



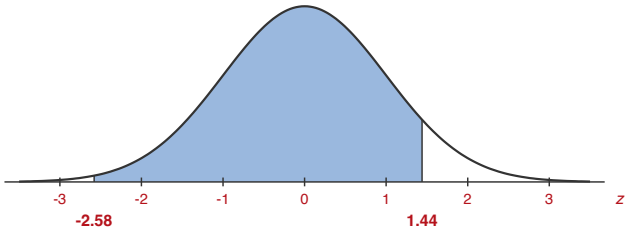
Row 1.4, column .02: left tail area $= 0.9222$
Right tail area $= 1.0 - 0.9222 = 0.0778$

12. Find the area for z between -1.96 and 1.96



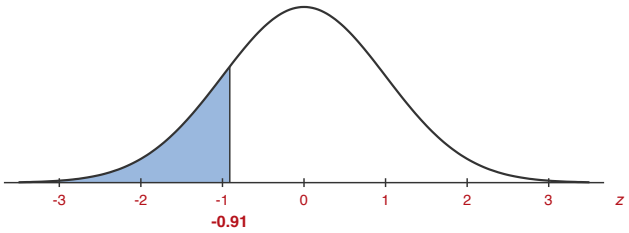
$z = 1.96$: row 1.9, column .06 gives 0.9750
$z = -1.96$: row -1.9, column .06 gives 0.0250
Area between $= 0.9750 - 0.0250 = 0.9500$

13. Find the area for z greater than -2.58 and less than 1.44



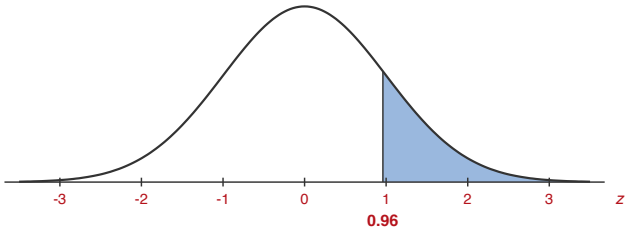
$z = 1.44$: row 1.4, column .04 gives 0.9251
$z = -2.58$: row -2.5, column .08 gives 0.0049
Area between $= 0.9251 - 0.0049 = 0.9202$

14. Find the area for z less than -0.91



Row -0.9, column .01: left tail area $= 0.1814$

15. Find the area where $z > 0.96$



Row 0.9, column .06: left tail area $= 0.8315$
Right tail area $= 1.0 - 0.8315 = 0.1685$